

Domain Control and Active Directory Lab

Part 1 – Creating and deploying Windows server

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Technologies Used

- Oracle VirtualBox
- Windows Server

Skills Demonstrated

- Windows Server Deployment
- Virtualization
- Enterprise Lab Design
- Initial Server Configuration.

Architecture

Host Machine

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----- Oracle VirtualBox

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----- DC1 (Windows Server, Future Domain Controllers)

OBJECTIVE:

Deploy a Windows Server 2022 virtual machine in Oracle VirtualBox that will serve as the foundation for an enterprise Active Directory environment. This phase focuses on creating the Domain Controller virtual machine, installing Windows Server, and preparing the server for future configuration and promotion to a Domain Controller.

Lab Setup

Windows server 2022 virtual machine was created in Oracle Virtual box, with Windows Server 2022 ISO, VM was named as DC1, in case additional Domain Controllers are added to environment. This Virtual Machine (VM) will be used to contain the AD and DC network, which will be used to manage user accounts, permissions, and network access.

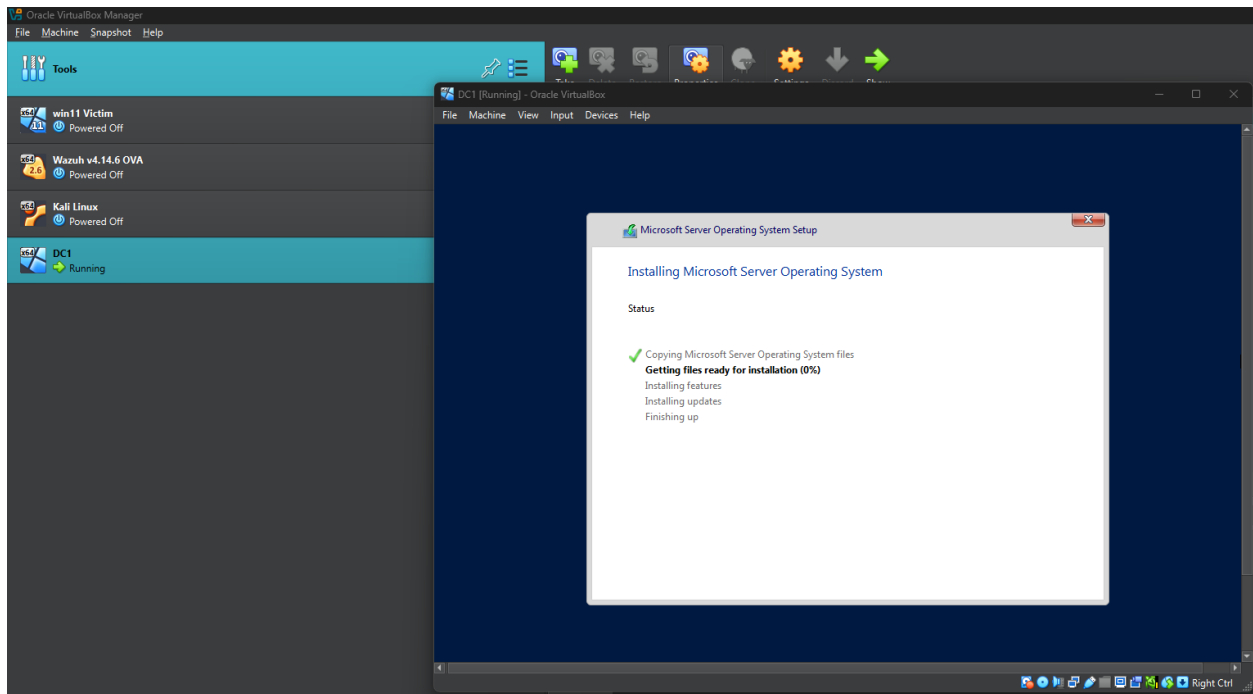


Figure 1. Windows Server, installation in Oracle VirtualBox. This virtual machine (DC1) will later be configured as the organization's Domain Controller hosting Active Directory Domain Services.



Figure 2. Windows Server 2022 successfully has been installed and deployed on DC1 virtual machine.

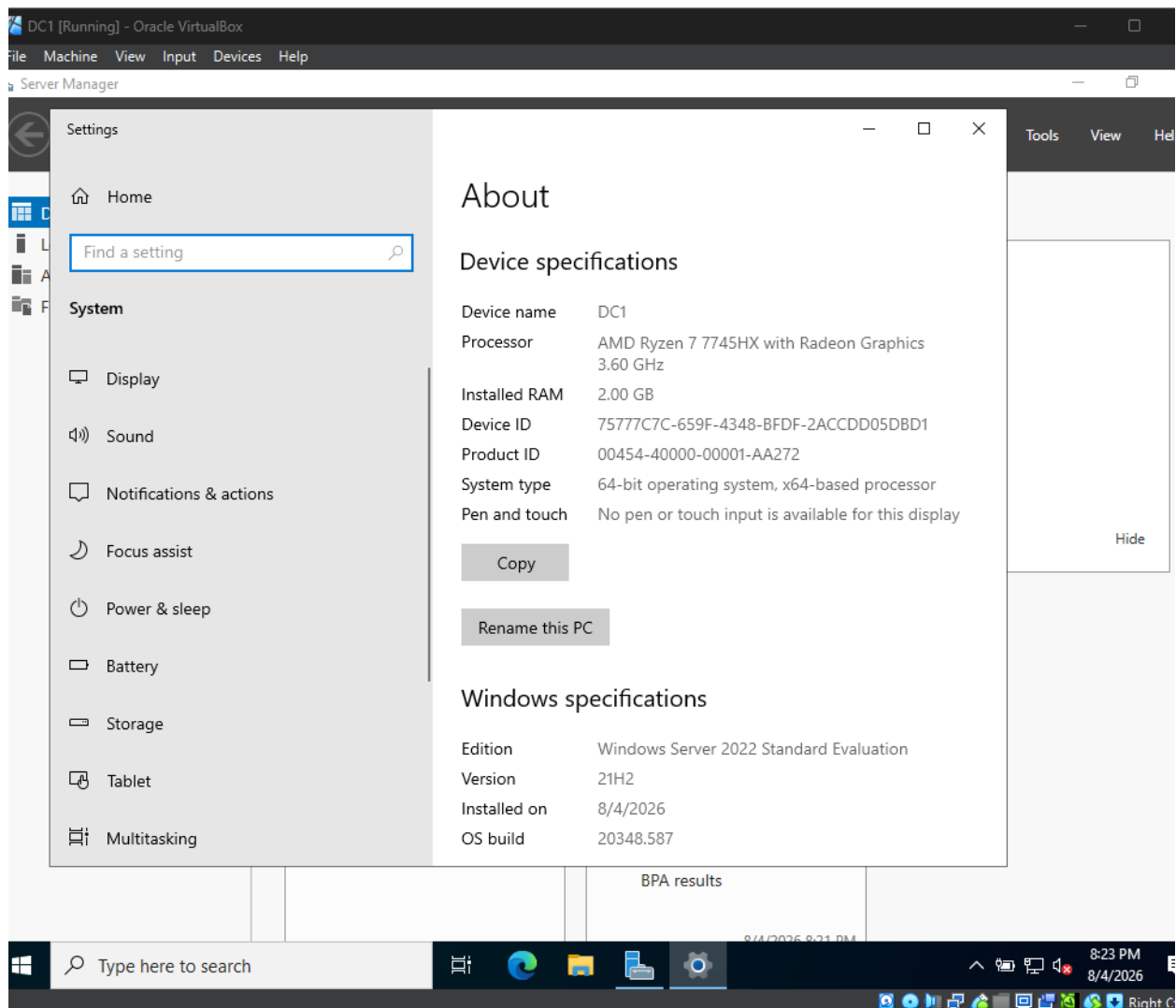


Figure 3. Windows Server was successfully renamed to DC1 to establish a standardized hostname for future Active Directory deployment.

Lessons Learned

- Successfully deployed a Windows Server 2022 virtual machine.
- Practiced enterprise virtualization using Oracle VirtualBox.
- Established a consistent server naming convention (DC1).
- Prepared the server for networking configuration and Active Directory installation.

Next Steps

- Configure static IPv4 addressing.
- Configure DNS settings.
- Rename network adapters.
- Verify network connectivity.
- Prepare for Active Directory Domain Services installation.